

THEMED PANELS

Three types of color-coded panels are used to present a more detailed focus on selected subjects. These panels appear both on explanatory pages and in feature profiles.

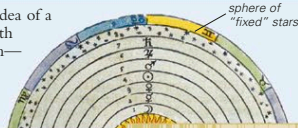
MYTHS AND STORIES

As well as being studied scientifically, objects in the night sky have featured in myths, superstitions, and folklore, which form the subject of this type of panel.

EXPLORING SPACE

ARISTOTLE'S SPHERES

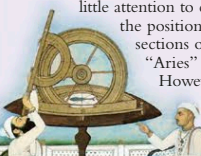
Until the 17th century CE, the idea of a celestial sphere surrounding Earth was not just a convenient fiction—many people believed it had a physical reality. Such beliefs date back to a model of the universe developed by the



MYTHS AND STORIES

ASTROLOGY AND THE ECLIPTIC

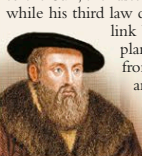
Astrology is the study of the positions and movements of the Sun, Moon, and planets in the sky in the belief that these influence human affairs. At one time, when astronomy was applied mainly to devising calendars, astronomy and astrology were intertwined, but their aims and methods have now diverged. Astrologers pay little attention to constellations, but measure the positions of the Sun and planets in sections of the ecliptic that they call "Aries" and "Taurus," for example. However, these sections no longer match the constellations of Aries, Taurus, and so on.



STARGAZER

JOHANNES KEPLER

The German astronomer Johannes Kepler (1571–1630) discovered the laws of planetary motion. His first law states that planets orbit the Sun in elliptical paths. The next states that the closer a planet comes to the Sun, the faster it moves, while his third law describes the link between a planet's distance from the Sun and its orbital period.



Newton used Kepler's laws to

EXPLORING SPACE

This type of feature is used to describe the study of space either from Earth's surface or from spacecraft. Individual panels describe particular discoveries or investigations.

BIOGRAPHY

Profiles of notable astronomers and pioneers of spaceflight, as well as a brief summary of their achievements, appear in this type of panel.

CONTRIBUTORS

Martin Rees General editor

Robert Dinwiddie What Is the Universe?, The Beginning and End of the Universe, The View From Earth, The Solar System, The Milky Way

Philip Eales The Milky Way

David Hughes The Solar System

Iain Nicolson Glossary

Ian Ridpath The View From Earth, The Night Sky

Robin Scagell The View from Earth

Giles Sparrow The Solar System, Beyond the Milky Way

Pam Spence The Milky Way

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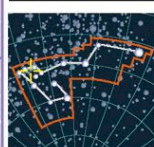
Kevin Tildsley The Milky Way

David Rothery The Solar System

name or astronomical catalog number of feature (features without a popular name are identified by number)

EMISSION NEBULA

Carina Nebula



CARINA

CATALOG NUMBER
NGC 3372
DISTANCE FROM SUN
8,000 light-years
MAGNITUDE 1

locator map shows constellation in which feature can be found and its position within the constellation

table of summary information (varies between sections)

selected features are described in double-page feature profiles

FEATURE PROFILES

Throughout the Guide to the Universe, introductory pages are often followed by profiles of a selection of specific objects. For example, the introduction to star formation (left) is followed by profiles of actual star-forming regions in the Milky Way (above).

themed panel (see above)

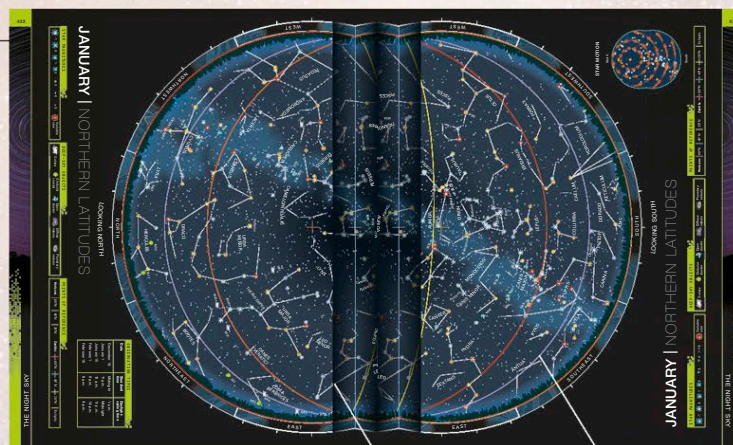
BEYOND THE MILKY WAY

This section looks at features found beyond our own galaxy, including other galaxies and galaxy clusters and superclusters, the largest known structures in the universe.

THE GREEK ALPHABET

Astronomers use a convention for naming some stars in which Greek letters are assigned to stars according to their brightness. These letters appear on some of the charts in this book.

α alpha	ν nu
β beta	ξ xi
γ gamma	ο omicron
δ delta	π pi
ε epsilon	ρ rho
ζ zeta	σ sigma
η eta	τ tau
θ theta	υ upsilon
ι iota	φ phi
κ kappa	χ chi
λ lambda	ψ psi
μ mu	ω omega



MONTHLY SKY GUIDE

This section includes two charts for each month of the year for observers in northern and southern latitudes. The section is described in more detail on pp.428–429.

chart on this page shows view looking north, with view to south on facing page

lines on chart show reference points for observers at different latitudes

locator shows constellation in context

artwork of figure depicted by constellation